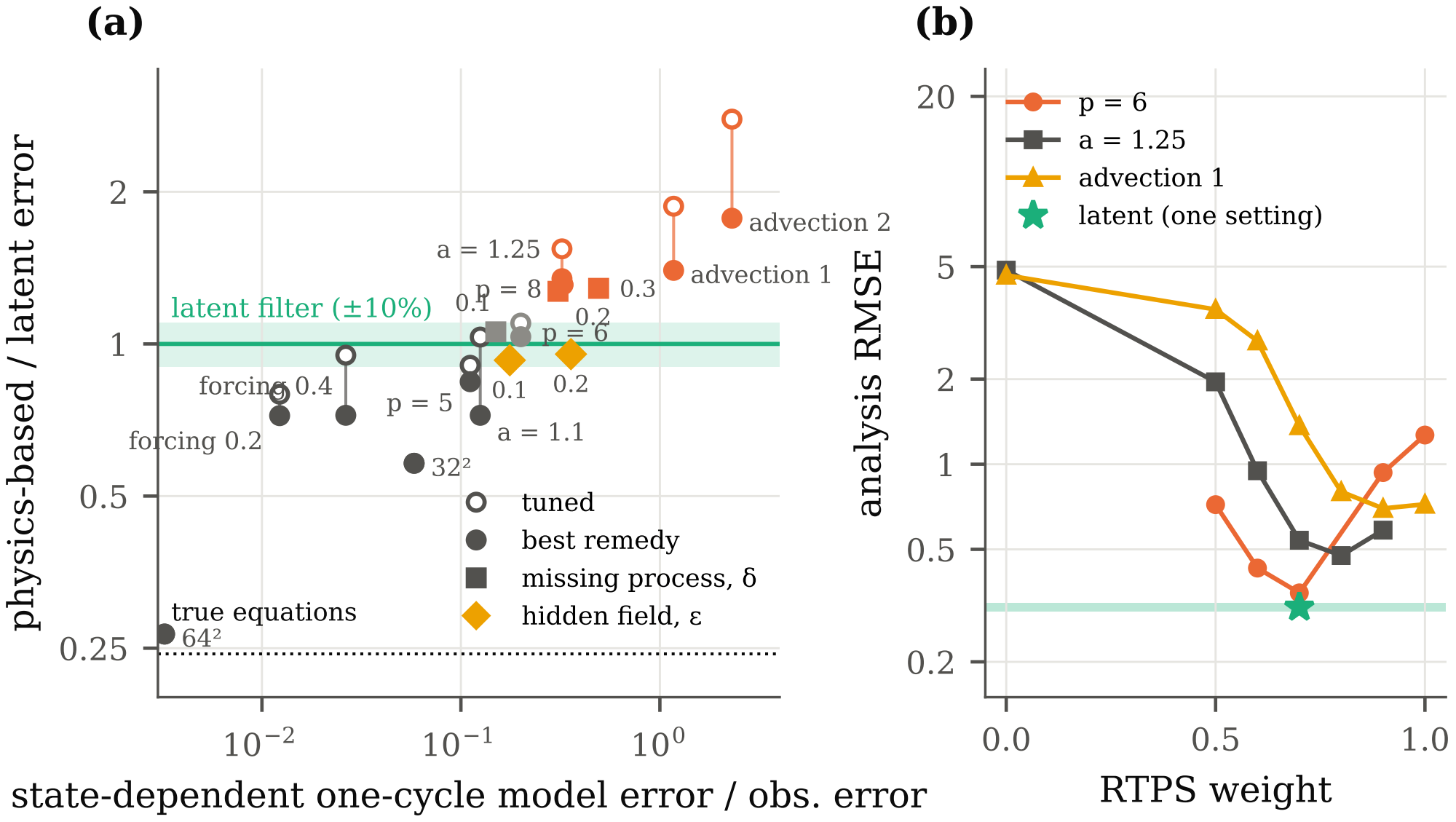
(a)
(b)
physics-based / latent error
state-dependent one-cycle model error / obs. error
latent filter (±10%)
advection 2
advection 1
a = 1.25
p = 8
0.3
0.1
0.2
p = 6
forcing 0.4
forcing 0.2
p = 5
0.1
0.2
a = 1.1
32²
tuned
best remedy
missing process, δ
hidden field, ε
true equations
64²
2
1
0.5
0.25
10⁻²
10⁻¹
10⁰
analysis RMSE
RTPS weight
p = 6
a = 1.25
advection 1
latent (one setting)
20
5
2
1
0.5
0.2
0.0
0.5
1.0